

NETHMI NIMSARANI THAMBAWITA

ASSOCIATE SOFTWARE ENGINEER | FULL-STACK DEVELOPER

Dehiwala, Sri Lanka | +94 71 336 5646 | thambawitanethmi@gmail.com

LinkedIn: [Nethmi Thambawita](#) | **GitHub:** [NethmiThambawita](#)

PROFESSIONAL SUMMARY

Associate Software Engineer with 1.5+ years of experience building full-stack web and mobile solutions for enterprise clients in banking and academic research. Delivered a real-time call center reporting dashboard processing 60,000+ row datasets and a cross-platform React Native field-data application with offline-first architecture and biometric authentication for a university research platform. Proficient across the MERN stack, .NET Core, and RESTful API design, with hands-on experience in API security testing, QA/UAT methodologies (Postman, SoapUI), and technical documentation (SRS, SOW, design specs) for client-facing engagements. Co-authored an IEEE-published research paper on device anomaly detection, and brings strong problem-solving and collaboration skills to cross-functional Agile delivery teams.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, C#, HTML, CSS, php, Java, Python
 - **Frontend:** React.js, React Native
 - **Backend:** Node.js, Express.js, .NET Core, Spring Boot, REST APIs
 - **Databases:** SSMS (SQL Server), MongoDB, MySQL
 - **Tools & DevOps:** Git, Docker, Figma, Postman
 - **Testing:** Postman, SoapUI, Cypress, UAT/QA methodology
 - **Data & Reporting:** Python (pandas, openpyxl), ClosedXML, Excel/CSV automation
-

WORK EXPERIENCE

DMS Electronics (Pvt) Ltd.

July 2025 – Present

Associate Software Engineer | Full Time | On Site

- Proactively analyze customer problems in a contact center and deliver effective solutions to enhance customer experience.
- Built an Admin dashboard application using React.js, Node.js, TypeScript, and MongoDB to manage user access accounts, enhancing usability and control.
- Designed and built a mobile application and wireframes for a research data collection platform using the MERN stack and Figma, focusing on a seamless and user-centric experience.
- Identified and resolved security vulnerabilities in the HSM Authentication Server to meet IT compliance requirements and conducted end-to-end QA testing of password authentication APIs, validating allowed and restricted character set handling against security specifications.
- Participated in daily stand-ups, sprint planning, and code reviews within a cross-functional Agile team, partnering with QA to ensure consistent on-time feature delivery.
- Managed end-to-end project activities, including client demonstrations, progress updates, and stakeholder communication throughout the implementation lifecycle.
- Actively contributed to preparing and reviewing key project documentation, including Software Requirement Specifications (SRS), Design Documents, and Statements of Work (SOW).
- Contributed to Asterisk — recreation of a web application combined with an agent call management system (Linux-based).
- Resolved client and call center system issues through troubleshooting and root cause analysis.

- Proactively analyzed customer problems in a contact center and delivered effective solutions to enhance customer experience.
- Built an Admin dashboard application using React.js, Node.js, TypeScript, and MongoDB to manage user access accounts, enhancing usability and control.
- Designed and built a mobile application and wireframes for a research data collection platform using the MERN stack and Figma, focusing on a seamless and user-centric experience.

Project: Research Data Collection Application*Mobile App Development | DMS Electronics*

- **Mobile Application Development:** Designed and developed a cross-platform mobile application using React Native for bird, bivalve, and phenology field surveys, enabling efficient and user-friendly research data collection.
- **Authentication & Security:** Implemented secure user authentication with Email, Google Sign-In, and Biometric (Fingerprint) authentication to provide flexible and secure access.
- **Dashboard & Data Management:** Developed a dynamic dashboard with advanced search and filtering capabilities based on date, location, and observation count for efficient data management and navigation.
- **Offline-First Architecture:** Implemented offline data collection using SQLite local storage, allowing uninterrupted field operations in low or no network connectivity environments.
- **Data Synchronization:** Developed automatic background synchronization to securely sync locally stored data with the backend once network connectivity was restored.
- **Location Services:** Integrated GPS-based location capture and validation to ensure accurate, location-based field observations.
- **User Experience:** Implemented automatic Light/Dark mode adaptation and optimized application performance for handling large datasets while maintaining a responsive user experience.

Technologies: React Native, React Native Paper, Node.js, Express.js, MongoDB, SQLite, Google

Authentication, Biometric Authentication, GPS Services, REST APIs, Docker

Call Detail Reporting Application*Website Development | DMS Electronics*

- Integrated four SQL Server databases with automatic primary/fallback failover to ensure high availability of reporting data.
- Developed 12 report modules covering service metrics, IVR flow analysis, agent performance, and customer satisfaction (CSAT).
- Built a real-time dashboard with live KPI polling (60s intervals) displaying call volume, service level, and abandon rate.
- Implemented role-based access control at both UI and controller layers, with BCrypt password hashing and secure cookie authentication.
- Engineered memory-efficient Excel/CSV export handling 60,000+ row datasets using ClosedXML with optimized styling for large outputs.
- Designed flexible stored procedure schema mapping to handle multi-variant column names across database versions without breaking reports.
- Applied date validation, row-limit safeguards, and timeout retry logic across all report endpoints to prevent performance degradation.

Technologies: Razor Views (ASP.NET MVC) , ASP.NET Core MVC, C#, .NET 4.8 , Microsoft SQL Server, IIS(Hosting).

PROJECTS

1. **AI-Driven Device Behavior Anomaly Detection System with Predictive Lost Item Risk Analytics**

Website Development | SLIIT University Final Year Research Project | IEEE Published

- Developed a web-based device tracking and monitoring system capable of detecting anomalous device movements without user interaction and generating real-time alerts to help prevent loss or theft.
- Designed and integrated a machine learning model to identify high-risk locations within a university environment where device loss is more likely to occur, enabling predictive risk analysis.
- Implemented real-time data processing and visualization features to improve device monitoring, security awareness, and decision-making.
- Contributed across the full development lifecycle — system design, frontend/backend implementation, testing, and deployment — within a collaborative team environment.

Technologies: React.js, Node.js, MongoDB, Python/flask

Link - <https://github.com/Minukweerakoon/Reclaim->

2. **Zero Waste Web Application**

Website Development | Group Project | SLIIT

- Developed a web-based waste management platform that connects citizens, waste collection services, and recycling centers through a centralized system.
- Implemented features to streamline waste collection requests, recycling coordination, and stakeholder communication.
- Designed a user-friendly interface to encourage responsible waste disposal and improve operational efficiency.
- Collaborated in an Agile team environment, contributing to system design, development, testing, and project management activities.

Technologies: React.js(vite), Node.js, Express.js, MongoDB, Tailwind CSS

Link - <https://github.com/NethmiThambawita/ZeroWasteHub-ITPM>

3. **PAWS Pet Care Web Application**

Website Development | Group Project | SLIIT

- Developed a full-stack web application that enables customers to schedule appointments for animal purchases, book purchase dates, create animal-related events, and purchase pet products online.
- **Designed and developed the Pet Shop module**, allowing customers to purchase pet essentials while providing administrators with tools to manage products, inventory, orders, and sales to support the shop's operational expenses.
- Implemented an administrative dashboard for managing animal listings, product inventory, appointments, customer interactions, and pet shop operations.

- Designed a responsive and user-friendly interface to enhance customer experience and streamline administrative operations.
- Contributed to both frontend and backend development, working across the React.js client and Node.js/Express API layers.

Technologies: MongoDB, Express.js, React.js, Node.js (MERN), CSS

Link - <https://github.com/NethmiThambawita/PawsPetCare>

EDUCATION

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY (SLIIT) 2022 – 2026

- BSc (Hons) in Information Technology Specializing in Information Technology

SCHOOL: DHARMASHOKA COLLEGE, AMBALANGODA

- G.C.E ORDINARY LEVEL: 2017
 - G.C.E ADVANCED LEVEL in The Stream of Physical Science: 2021
-

OTHER SKILLS

- Good understanding of Software Development Life Cycle (SDLC), Agile methodologies, and Scrum practices.
 - Version control using Git, GitHub
-

REFERENCE

Mr. Nipul Goonesekara

Telecommunication Technologist

United Nations Mission in South Sudan (UNMISS) – FTS
Malakal, South Sudan

+211 912 167 452

Mrs. Aruni Ginige

Civil Engineer

Provincial Health Engineer Office, First Floor, Ossusala
Building, Wakwella Road, Galle

+94 71 837 6412